Hearing on the Move: A Review of Acoustic Perception for Autonomous Intelligent Systems

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Abstract

Autonomous systems that hear can detect and locate events that cameras and lidar miss, such as a siren behind a building or a voice under rubble, yet audition remains the least developed sense on robots, drones and vehicles. This review surveys acoustic perception for these three platform families across six tasks: localization and tracking, event detection, separation and enhancement, joint localization and detection, acoustic mapping, and audio-driven embodied behavior, tracing each from array signal processing to learned, multimodal and language-grounded models. We argue that one problem sets the field apart from audio on static devices: ego-noise*, the sound a platform makes itself. When it dominates, methods that model it explicitly, especially with the platform's own motor or joint state, outperform those that treat it as generic interference. We review datasets, simulators, toolkits and evaluation protocols, and report reproducible case studies on three public datasets. On a drone, a learned rotor-noise model raises in-flight localization accuracy within* 10° *from 18% to 60%, and to 70% with motor-speed bins, and it gives about 10 dB of beamforming gain. A Wiener filter with one noise model reaches 71% without motor data, so the added value of the motor state is smaller than it first appears. On a second drone, recorded speech fails about 10 dB earlier than white noise. Across five drones, the tonal share of ego-noise ranges from 1% to 70%. The highest values come from drones that were not flying, and rotor-speed variation in flight lowers the tonal share. A learned spatial model removes 6 to 13 dB on every drone. On a robot-head array and a spherical array at high SNR, whitening with the same kind of model lowers accuracy in six of eight comparisons, whereas a Wiener filter does not. This marks where such models stop helping. We close with open problems, from hearing at very low signal-to-noise ratios and vibration-referenced noise cancellation to transfer across platforms and privacy.*

1. Introduction

Sound gives an autonomous system a sense that works around corners, in the dark, through smoke and in every direction at once. Cameras and lidar need line of sight and a field of view; microphones need neither. A vehicle can hear another vehicle approaching a blind corner more than a second before a visual detector sees it [79]. A drone can search for a voice it cannot see [20, 48]. An embodied agent finds a sounding object in an unexplored house faster when it listens as well as looks [9, 32], and a robot arm can complete a task inside an opaque bag by listening to the contact [24].

Despite this, hearing remains the least developed sense on most autonomous platforms. The obstacle is not a lack of methods. Microphone-array processing is mature [7, 49, 78], large pretrained models recognize hundreds of sound classes [16, 37, 38, 50], and speech recognition has become robust to many real-world conditions [71]. The obstacle is the platform itself. Robots, drones and vehicles move, vibrate and, above all, make noise. On a quadrotor, the rotor noise at the drone's own microphones is typically far louder than the sound it is trying to hear [74, 84], and on a humanoid, the noise of its own joint motors corrupts what it records whenever it moves [98].

Robot audition emerged as a research field in 2000, when Nakadai *et al*. proposed *active audition*: a robot that listens while it moves instead of stopping to perceive [62]. The field produced the HARK toolkit [63], real-time localization and tracking on mobile robots [88, 89], and later ODAS for embedded hardware [41]. Since then, three shifts have changed it. First, drones became a major platform with an extreme ego-noise problem [17, 58, 84, 95]. Second, learned models moved from niche to default, including joint detection and localization [1, 69, 81]. Third, audio became one input among several to embodied and multimodal intelligence, driven largely by the vision and learning communities [9, 34, 51, 57].

Survey	Rob.	Drone	Veh.	Ego-n.	Learn.	Embod.
Argentieri *et al*. [4]	✓			partial		
Rascon, Meza [72]	✓			partial		
Grumiaux *et al*. [42]					✓	
Mesaros *et al*. [61]					✓	
Chevtchenko *et al*. [17]		✓		partial		
This review	✓	✓	✓	✓	✓	✓

Table 1. Scope of related surveys: platforms (robots, drones, vehicles), treatment of ego-noise, learned methods, and embodied or multimodal learning.

Relation to prior surveys. Table 1 compares this review with earlier ones. Two surveys mapped localization for robots up to 2017 [4, 72]; Grumiaux *et al*. reviewed deep learning for localization in rooms [42]; Mesaros *et al*. gave a tutorial on sound event detection [61]; and Chevtchenko *et al*. systematically reviewed drone-based localization, finding that only three of 49 studies implemented real-time on-board localization [17]. To our knowledge, none treats ego-noise across platforms, covers the embodied and audio-visual work of the vision and learning communities, or links the two.

Scope. Our scope is acoustic perception on board a moving, self-powered platform; fixed arrays and smart speakers are included only where their methods transfer, and underwater sonar is excluded.

Contributions.

1. A task-centered map of the field (Secs. 4 to 7 and 9) that links classical and learned methods for each task.
2. A cross-platform treatment of ego-noise (Sec. 8) that organizes suppression methods by the information they exploit.
3. A review of audio in embodied, audio-visual and language-grounded intelligence (Sec. 10).
4. A guide to datasets, simulators, toolkits (Sec. 11) and evaluation protocols (Sec. 12).
5. Reproducible case studies on three public datasets (Sec. 13): in-flight localization on a drone with block-bootstrap statistics and controls, recorded speech on a second drone, ego-noise structure across five drones, and localization with a robot-head and a spherical array. Together they quantify when an explicit noise model and motor-speed information help, and when they do not.
6. Open problems with concrete directions (Sec. 14).

Figure 1 summarizes the structure, and Fig. 2 places key milestones in time.

2. Background and Problem Formulation

Signal model. An array of M microphones at positions $\mathbf{r}_1, \ldots, \mathbf{r}_M$ observes, in the short-time Fourier transform (STFT) domain,

$$\mathbf{x}(f,t) = \sum_{k=1}^{K} \mathbf{a}(f, \mathbf{u}_k)\, s_k(f,t) + \mathbf{n}_{\text{ego}}(f,t) + \mathbf{n}_{\text{env}}(f,t), \tag{1}$$

where s_k are K target sources, $\mathbf{n}_{\text{ego}}$ is the noise the platform makes itself, and $\mathbf{n}_{\text{env}}$ is environmental noise and reverberation. For a far-field source from unit direction $\mathbf{u}$, the steering vector is

$$\mathbf{a}(f, \mathbf{u}) = \left[e^{j2\pi f\, \mathbf{r}_1^\top \mathbf{u}/c}, \ldots, e^{j2\pi f\, \mathbf{r}_M^\top \mathbf{u}/c} \right]^\top, \tag{2}$$

with c the speed of sound. Spatial covariances $\mathbf{R}_x(f) = \mathbb{E}[\mathbf{x}\mathbf{x}^H]$ and $\mathbf{R}_n(f)$ for the noise summarize the second-order statistics that most classical methods use.

What makes autonomous platforms different. Three properties of Eq. (1) distinguish autonomous platforms from static devices. First, $\mathbf{n}_{\text{ego}}$ is often the dominant term, so the signal-to-noise ratio (SNR) at the microphones is strongly negative [74, 84]. Second, the array moves, so $\mathbf{u}_k$ changes over time even for static sources, and reverberation changes as the platform moves through a space. Third, $\mathbf{n}_{\text{ego}}$ is a function of the platform's own state, such as rotor speeds or joint velocities, which the platform knows.

Binaural hearing. Two-microphone robots, such as humanoid heads, rely on interaural time and level differences, as human hearing does. They suffer from front-back ambiguity, which head movement can resolve [56]; this was one of the original motivations for active audition [62].

Tasks. We use the following task definitions. *Localization* estimates the direction (and sometimes distance) of each source; *tracking* links estimates over time. *Sound event detection* (SED) estimates which classes are active and when [61]. *Separation and enhancement* recover the target signals s_k. *Sound event localization and detection* (SELD) estimates class, activity and direction jointly [1]. *Acoustic mapping* recovers the positions of sources, of the platform, or of the surrounding geometry.

3. Platforms and Their Acoustic Conditions

The same algorithm can succeed on a smart speaker and fail on a drone, because the platform decides what the microphones hear. Table 2 compares the three platform families along the dimensions that matter most.

Ground robots and humanoids. Robot audition grew out of humanoid robotics, where the goal was natural interaction: localizing and separating several talkers and recognizing

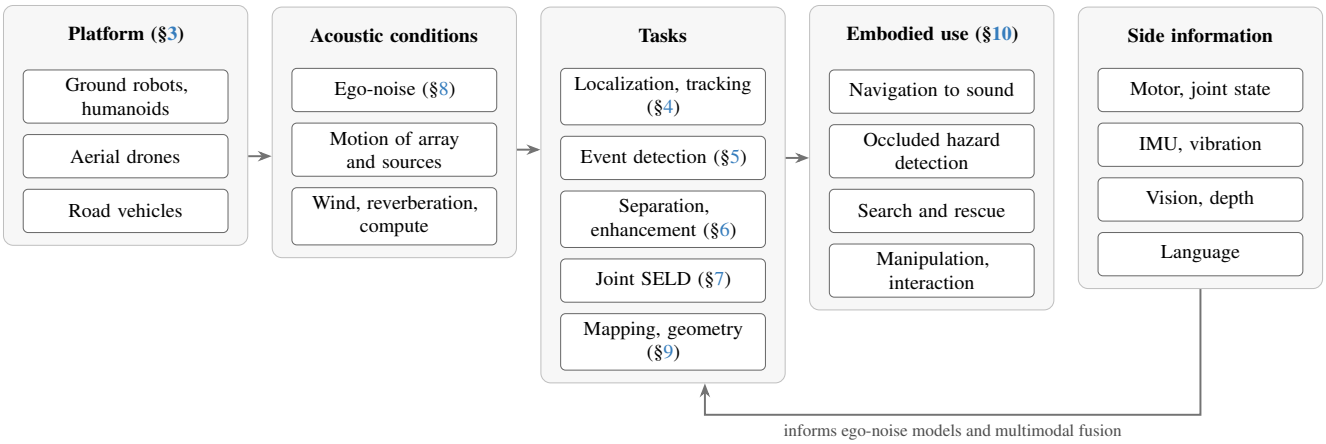


Figure 1. Structure of the review. A platform sets the acoustic conditions, of which ego-noise is the most severe. Perception tasks turn microphone signals into events, directions and geometry that embodied agents act on. Side information that only an autonomous platform has (its own motor, motion and visual state) can inform ego-noise models and multimodal fusion.

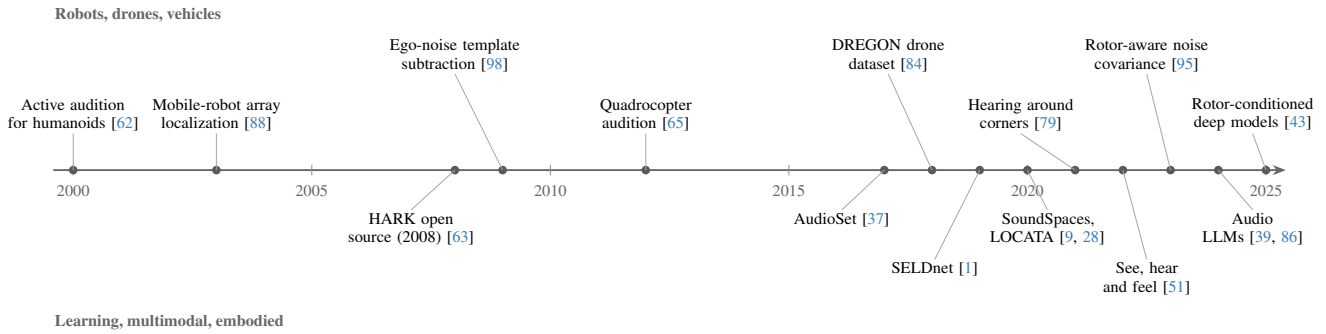


Figure 2. Selected milestones in acoustic perception for autonomous systems. Above the axis: platform-centered robot, drone and vehicle audition. Below: learning, audio-visual and embodied work. Positions follow the publication year.

their speech [62, 63]. Mobile robots added real-time localization and tracking with on-board arrays [40, 88, 89]. Arrays are small (a head or a torso), the robot often moves, and the dominant ego-noise comes from fans and from joint motors that switch on and off with each gesture or step [76, 98]. Because a robot knows its own joint state, this noise is partly predictable, which has made robots the proving ground for motor-informed ego-noise suppression [75, 77, 98]. Robot manipulators increasingly use sound as a contact sensor, for example to estimate liquid height while pouring [52] or to act inside occluded containers [24].

Aerial drones. Drones are the hardest case. Microphones sit tens of centimeters from rotors spinning thousands of times per minute, and the rotor noise combines strong harmonics of each rotor's shaft rate with broadband aerodynamic noise [84, 95]. Recordings of drone ego-noise have very low SNR, change with the drone's maneuvers, and are highly correlated across microphones [74]. Drone audition is motivated mainly by search and rescue: 82% of the 49 studies in a recent systematic review target this use [17], and fielded systems have localized voices from the air in outdoor demonstrations [48]. Early work localized sources outdoors with an array embedded in a quadrocopter [65] and learned the rotor-noise correlation matrix to suppress it [30]. Public datasets now provide on-board recordings with ground truth [74, 84, 94]. Payload and power budgets limit both the array and the on-board compute.

Road vehicles. Vehicles use sound mainly for safety: detecting emergency vehicles and hazards that are not yet visible. Early work modeled the urban soundscape to detect horns and sirens as anomalies [60], and later work localized and classified sirens from stereo recordings by segmenting spectrograms [59]. A roof-mounted array can detect a vehicle approaching a blind corner from reflections off nearby walls, before it enters line of sight [79], and even two low-cost microphones can detect sirens and estimate their direction [85]. Self-supervised learning from stereo sound has been used to track moving vehicles, with a visual network as

	Ground robots	Drones	Vehicles
Main use	interaction, speech, manipulation	search and rescue	safety, occlusion
Ego-noise	fans, joint motors	rotors (tonal and broadband)	engine, tires, wind
Noise varies with	joint motion	rotor speed	speed, road
Side info	joint state	motor speed, IMU	speed, CAN data
Array size	head or torso	limited by payload	roof, large
Key refs	[62, 77, 89, 98]	[30, 48, 84, 95]	[31, 59, 60, 79]

Table 2. Acoustic conditions on the three platform families. In every case the platform's own state carries information about its noise, which static devices do not have.

teacher [31]. The main noise sources are the vehicle's own engine and tires and, at speed, wind at the microphones.

4. Sound Source Localization and Tracking

Localization is the most studied task in robot audition and the one most affected by ego-noise. Table 3 summarizes representative methods.

4.1. Classical methods

Generalized cross-correlation with phase transform (GCC-PHAT) estimates the time delay between each pair of microphones [49]. Steered response power with phase transform (SRP-PHAT) sums these correlations over all pairs for every candidate direction and picks the maximum [23]. Subspace methods such as MUSIC [78] split $\mathbf{R}_x(f)$ into signal and noise subspaces and search for steering vectors (Eq. (2)) orthogonal to the latter. Both assume that noise is spatially white, which ego-noise is not (Sec. 8). Generalized eigenvalue decomposition MUSIC (GEVD-MUSIC) removes this assumption by whitening with a noise covariance $\mathbf{R}_n(f)$, solving

$$\mathbf{R}_x(f)\,\mathbf{e} = \lambda\,\mathbf{R}_n(f)\,\mathbf{e}; \tag{3}$$

it improved robot localization robustness by 5 to 6 dB SNR [64] and became the basis of drone audition [30, 65].

On mobile robots, Valin *et al*. localized sources with an eight-microphone array in real time [88] and then combined a frequency-domain steered beamformer with a particle filter to track several moving sources over about 7 m [89]. Later work reduced the cost of SRP-PHAT by coarse-to-fine scanning and tracked directions with a modified Kalman filter, using up to four and thirty times less computation respectively [40]; these methods underlie the ODAS toolkit [41].

4.2. Binaural and active localization

Two-microphone systems use interaural cues. Deep networks that map the full cross-correlation function and interaural level differences to azimuth, combined with head rotation to resolve front-back confusions, localize multiple sources robustly in reverberant rooms [56]. More generally, an agent can move to hear better: active audition listens during motion rather than in stop-perceive-act cycles [62], and Move2Hear learns where to move to improve source separation [57].

4.3. Learned methods

Neural localizers learn the map from multichannel features (phase maps, GCC features, SRP maps or raw spectrograms) to directions, by classification over a grid or by regression. A convolutional network trained on noise signals generalizes to speech [8]; deep networks localize and detect multiple speakers from a robot [46]; and Cross3D tracks a source in strongly reverberant rooms by running a causal 3D convolutional network over SRP-PHAT maps, trained on simulated trajectories and evaluated on real LOCATA recordings [22]. Grumiaux *et al*. survey this literature in depth [42]. On drones, a neural time-frequency mask that suppresses ego-noise before SRP-based or spatial-filtering localization outperforms either component alone [93], and a TDOA-based method with noise angular spectrum subtraction works at low signal-to-drone-noise ratios on an irregular on-board array [58].

4.4. Tracking

Moving platforms need temporally consistent estimates. Particle filters [89] and Kalman filters [40] link frame-wise estimates and suppress isolated errors. The LOCATA challenge benchmarks localization and tracking from single static sources up to multiple moving sources observed by a moving array [28].

5. Sound Event Detection and Classification

Detection answers *what* is sounding and *when*; Mesaros *et al*. give a tutorial treatment of the task, its methods and its evaluation [61].

Data. Large labeled collections made modern detectors possible. ESC-50 provides 2,000 five-second clips over 50 environmental classes [68]; AudioSet provides a sound-event ontology and about two million human-labeled 10-second clips from web video [37]; FSD50K provides over 51k openly licensed clips labeled with 200 AudioSet classes [29]; and VGGSound provides about 200k web videos whose audio and visual content correspond [15].

Models. Pretrained audio neural networks (PANNs) trained on AudioSet transfer well to downstream tasks [50]. The Audio Spectrogram Transformer (AST) showed that

Method	Venue	Key idea	Platform
GCC-PHAT [49]	TASSP'76	pairwise delay	any
SRP-PHAT [23]	thesis'00	steered power	any
MUSIC [78]	TAP'86	signal subspace	any
Active audition [62]	AAAI'00	listen while moving	humanoid
Valin *et al*. [89]	RAS'07	beamformer + particle filter	mobile robot
GEVD-MUSIC [64]	IROS'11	noise whitening	robot
Okutani *et al*. [65]	IROS'12	GEVD on a quadrocopter	drone
Chakrabarty, Habets [8]	WASPAA '17	CNN on phase	static
Ma *et al*. [56]	TASLP'17	DNN + head motion	binaural head
He *et al*. [46]	ICRA'18	DNN, multi-speaker	robot
Cross3D [22]	TASLP'21	3D CNN on SRP maps	static
Wang, Cavallaro [93]	Sens. J'22	DNN mask + SRP	drone
Manamperi *et al*. [58]	TASLP'22	TDOA + noise subtraction	drone

Table 3. Representative sound source localization methods, ordered from classical to learned.

a convolution-free, attention-only model reaches state-of-the-art accuracy, 0.485 mAP on AudioSet [38]. BEATs learns general audio representations by self-supervised prediction of discrete labels from an acoustic tokenizer [16]. Contrastive language-audio pretraining (CLAP) aligns audio with text and enables zero-shot classification of classes never seen in training [25].

On autonomous platforms. These models are trained on clean web audio, while a platform hears targets at low SNR under its own noise; domain shift is therefore the central challenge. Vehicle work has addressed it with task-specific pipelines: modeling the urban soundscape to detect anomalous alerting sounds and then classifying them [60], treating spectrograms as images to segment sirens from background noise before classifying and localizing them [59], and detecting sirens and their direction with only two low-cost microphones [85]. Robot manipulators use detection-style models to infer physical state from sound, for example liquid height during pouring [52]. Few of these systems have been tested under the strong ego-noise of drones, and we are not aware of a public benchmark for multi-class event detection on board a drone.

6. Separation, Enhancement and Speech

Separation recovers the target signal itself, for recognition or for a human operator.

Spatial filtering. The minimum variance distortionless response (MVDR) beamformer [7] keeps the target direction undistorted and minimizes everything else:

$$\mathbf{w}_{\mathrm{MVDR}}(f) = \frac{\mathbf{R}_n^{-1}(f)\,\mathbf{a}(f,\mathbf{u})}{\mathbf{a}^H(f,\mathbf{u})\,\mathbf{R}_n^{-1}(f)\,\mathbf{a}(f,\mathbf{u})}. \tag{4}$$

Like GEVD-MUSIC (Eq. (3)), it needs the noise covariance $\mathbf{R}_n$, which is exactly what an ego-noise model provides. Neural mask estimation made such beamformers practical without explicit noise recordings: a network estimates time-frequency masks for speech and noise, which yield the covariances that define the beamformer [47].

Learned separation. Supervised separation with deep networks covers speech enhancement, multi-talker separation and dereverberation [91]. Conv-TasNet works directly on waveforms and surpassed ideal time-frequency masks on speech separation [55]. Separation quality is now usually reported as scale-invariant signal-to-distortion ratio (SI-SDR), introduced because the earlier SDR could be misleading [73].

Audio-visual separation. Vision tells a separator which source to extract. The Sound of Pixels separates sources by the image regions that produce them [97]; multisensory features learned from synchronization separate on-screen from off-screen sound [66]; Looking to Listen isolates a chosen speaker from a crowd using the speaker's face [26]; and audio-visual object embeddings learned without labels support separation, localization and active-speaker detection [2]. An agent can also move to improve separation: Move2Hear learns camera and microphone motion that maximizes predicted separation quality [57].

On drones. Drone speech enhancement must cope with ego-noise far louder than speech. Neural masks estimated by a single-channel network suppress ego-noise before spatial processing [93]; audio-visual recordings from a quadcopter provide baselines for localization and enhancement [94]; and conditioning end-to-end enhancement networks on rotor speed improves speech recovery [43].

Speech recognition. For human-robot interaction, enhancement is usually followed by recognition. Large weakly supervised models such as Whisper, trained on 680,000 hours of transcribed audio, generalize zero-shot to many conditions [71], but their robustness to strong ego-noise has not been characterized.

7. Joint Sound Event Localization and Detection

Sound event localization and detection (SELD) estimates class, activity and direction together, which is the natural output format for an autonomous system that must know both what is happening and where.

Methods. SELDnet, a convolutional recurrent network, introduced joint detection and direction-of-arrival regression for overlapping sources [1]. The activity-coupled Cartesian DOA (ACCDOA) representation merged the two outputs into a single vector per class, whose length encodes activity and whose direction encodes the source direction [81]. Event-independent networks assign overlapping events to separate output tracks [6].

Evaluation. The first SELD evaluation, run in DCASE 2019, ranked systems by localization and detection scored separately; joint metrics that require an event to be both correctly detected and close to its true direction changed the ranking of several systems [69]. STARSS23 provides real recordings with spatiotemporal annotations together with 360-degree video, and shows that visual cues reduce localization error [82].

Gap. All SELD benchmarks so far use static arrays without ego-noise. Moving SELD to robots and drones requires both new data and methods that model the platform's own noise.

8. Ego-Noise: The Defining Problem

Ego-noise is the sound a platform makes itself. It differs from the interference that room acoustics and audio benchmarks assume in four ways. It is *loud*: on drones, targets are routinely recorded at strongly negative SNR [20, 74, 84], and in our drone case study localization breaks down between -10 and -20 dB (Sec. 13.1). It is *close and fixed*: its sources sit at known places on the body, so it has strong spatial structure, and drone ego-noise is highly correlated across microphones [74]. It is *non-stationary*: it changes with every gesture, maneuver or acceleration. And it is *partly predictable*, because the platform knows its own motor and joint state. The last property is what an autonomous platform has that a static device does not, and the most effective methods exploit it.

Structure of rotor and motor noise. A rotor turning at f_0 revolutions per second produces tones at multiples of f_0, strongest at multiples of the blade passing frequency Bf_0 for B blades, plus broadband aerodynamic noise; both shift with speed [95]. Robot joint motors produce harmonic noise whose structure follows motor speed [77]. Our case studies confirm and qualify both points on real drone data: predicted rotor tones follow the measured motor speed closely, yet on most drones the noise is largely broadband, and the share of energy in tones ranges from 1% to 70% across five drones (Sec. 13.3), so frequency-domain masking alone cannot remove it.

Approach	Information used	Platforms	Refs
Spatial noise model	noise-only recordings	robots, drones	[30, 64, 65]
State-informed	joint or motor state	robots, drones	[75–77, 95, 98]
Learned	paired training data	drones	[43, 93]
Reference sensors	vibration with phase	open	–

Table 4. Ego-noise suppression grouped by the information it exploits. Methods further down use more platform-specific information and need more platform-specific data.

A taxonomy of suppression methods. We group ego-noise methods by the information they exploit (Tab. 4).

(i) Spatial noise models. Because ego-noise sources are fixed relative to the array, their spatial covariance $\mathbf{R}_n(f)$ can be learned from noise-only recordings and used to whiten the observations, as in GEVD-MUSIC [64, 65] or MVDR (Eq. (4)). Furukawa *et al*. estimated the noise correlation matrix for a multirotor drone and showed that it improves localization [30].

(ii) State-informed models. The platform's own state predicts its noise. On robots, template subtraction selects a pre-recorded noise template from the current joint status [98], and motor data can guide or regularize multichannel dictionaries of ego-noise [75–77]. On drones, rotor-noise-aware covariance estimation combines a multi-sensory estimate of rotor noise power with a model of the acoustic paths from rotors to microphones, and reports up to about 28 dB rotor-noise reduction on in-flight recordings [95].

(iii) Learned suppression. Neural networks estimate time-frequency masks that remove ego-noise before localization [93], in the spirit of mask-based beamforming [47], or are conditioned directly on rotor speed for end-to-end enhancement [43]. These methods can capture non-linear structure that covariance models miss, but they need training data from the same platform.

(iv) Reference sensors. Accelerometers or other vibration sensors close to the noise sources can, in principle, provide a reference signal with phase, which would allow tones to be subtracted rather than masked. This direction is largely open for autonomous platforms; flight-controller IMUs are usually logged at rates too low to capture the acoustic band (Sec. 13.1).

Vehicles. Ego-noise on road vehicles is less studied in the perception literature. Engine and tire noise are partly predictable from vehicle speed and engine state, and wind noise grows with speed; the occluded-vehicle detector of Schulz *et al*. reports its headline accuracy with a static ego-vehicle [79]. Transferring state-informed ego-noise models from robots and drones to vehicles is a clear gap.

Real-time constraints. Ego-noise suppression must run on the platform. Only three of 49 drone localization studies in a recent systematic review ran in real time on board [17], which makes efficient methods [40, 41] as important as accurate ones.

9. Acoustic Mapping, Echoes and Geometry

Sound also reveals the space around a platform, which matters most when vision is degraded.

Mapping from sources. Acoustic SLAM jointly estimates the trajectory of a moving array and a map of the surrounding sources from direction-of-arrival measurements [27].

Active echoes. A platform that emits sound can infer geometry from the echoes. BatVision predicts depth and spatial layout from binaural recordings of emitted chirps [18], and VisualEchoes learns spatial image representations through echolocation, which transfer to depth estimation and navigation [34].

Geometry from ambient and binaural sound. Audio-visual floorplan reconstruction uses sound to infer the layout of rooms the camera has not seen [70]. Binaural recordings can predict the semantic segmentation and depth of the surrounding scene and can be super-resolved spatially [90], and video can turn mono audio into binaural audio by exploiting the visible scene layout [33].

Acoustic rendering. Neural acoustic fields learn a continuous model of how sound propagates between any emitter and listener position in a scene [54]. Visual acoustic matching re-renders a recording to sound as if it were made in the room shown in an image [12], and novel-view acoustic synthesis predicts what a scene sounds like from a new viewpoint [14]. These models give an agent an internal model of room acoustics, which could in turn support prediction of what it should hear and hence anomaly detection.

10. Audio in Embodied and Multimodal Intelligence

Much of the recent progress in machine hearing for agents has come from the vision and learning communities, where sound is treated as one modality of an embodied agent rather than as a signal to be cleaned. Table 5 summarizes representative work.

Audio-visual navigation. SoundSpaces introduced navigation to a sounding target in realistic scanned 3D environments, with audio rendered by geometric acoustic simulation, and showed that audio improves embodied navigation [9]. Gan *et al*. proposed a parallel formulation with a visual mapper, a sound-localization module and a planner [32]. Follow-up work learned to set waypoints with an acoustic memory [11], extended the task to semantically meaningful sounding objects [10], and to moving sound sources amid distracting noise [96]. Simulation platforms have matured: SoundSpaces 2.0 renders continuous, configurable acoustics for visual-acoustic learning [13], and Sonicverse provides a multisensory household simulator for agents that see and hear [36]. These platforms make large-scale training possible, but they usually omit the agent's own ego-noise, which Secs. 8 and 13 show is the dominant difficulty on real platforms.

Self-supervised audio-visual learning. Correspondence between sight and sound is a strong free supervisory signal. Ambient sound supervises visual representation learning [67]; audio-visual correspondence localizes sounding objects in images [3, 80]; audio-visual events can be localized in time in unconstrained video [87]; and audio-visual objects can be discovered and tracked from unlabeled video [2]. For autonomous vehicles, a visual network has served as teacher for tracking vehicles from stereo sound [31], and binaural recordings have been used to predict semantic segmentation and depth [90]. Sound can also detect what vision cannot yet see: an acoustic detector anticipated vehicles behind blind corners more than a second before a visual detector reached the same accuracy [79].

Manipulation and interaction. Audio carries contact information. Fusing vision, audio from a contact microphone, and touch improves robotic manipulation on dense packing and pouring [51]; audio estimates liquid height during pouring [52]; audio helps imitation learning when the scene is visually occluded, reaching 70% success on extracting keys from a bag, which the authors report is 50% higher than a policy without audio [24]; and object-centric datasets pair visual, auditory and tactile representations of the same objects [35].

Audio and language. Contrastive language-audio pretraining links sounds to text and enables zero-shot classification [25]. Audio language models now cast audio tasks as text generation [21], and perceive and reason about general

Work	Venue	Contribution	Setting
SoundSpaces [9]	ECCV'20	audio-visual navigation	simulation
Gan *et al*. [32]	ICRA'20	mapper + sound + planner	simulation
Waypoints [11]	ICLR'21	acoustic memory	simulation
Semantic AV-Nav [10]	CVPR'21	semantic sounding goals	simulation
Move2Hear [57]	ICCV'21	move to separate	simulation
SoundSpaces 2.0 [13]	NeurIPS'22	continuous acoustics	simulation
Catch me if you hear me [96]	RA-L'23	moving sound sources	simulation
Sonicverse [36]	ICRA'23	multisensory household	simulation
See, Hear, Feel [51]	CoRL'22	vision, audio, touch	real robot
Play it by Ear [24]	RSS'22	audio under occlusion	real robot
Hearing around corners [79]	RA-L'21	occluded vehicles	real vehicle

Table 5. Representative embodied and multimodal work that uses sound. Most audio-visual navigation is still developed in simulation without ego-noise.

sounds, speech and music [39, 86]. For autonomous systems, these models promise open-vocabulary event detection and natural-language reports of what the platform hears. Their robustness to ego-noise and their latency on embedded hardware are, as far as we are aware, untested.

11. Datasets, Simulators and Toolkits

Progress in acoustic perception for autonomous systems is limited less by models than by data recorded on real platforms. Table 6 lists the main public resources and what each can test.

Three gaps stand out. First, the benchmarks with rich annotations (STARSS23 for SELD, SoundSpaces for navigation, AudioSet for classification) do not include ego-noise, while the datasets that do (DREGON, AIRA-UAS, AVQ) are small and focus on localization and enhancement. Second, no public benchmark yet combines a moving platform, real ego-noise, the platform's state, high-rate vibration sensing, multiple sound classes and visual input. Such a benchmark would let the embodied-learning and robot-audition communities measure progress on the same problem. Third, availability is fragile: of the ten drone ego-noise sets released with the SP Cup, two could no longer be downloaded at the time of writing, so archival hosting with persistent identifiers matters as much as collection. Toolkits such as HARK and ODAS lower the barrier to real-time deployment, but they implement classical pipelines and would benefit from integration with learned models.

12. Evaluation Protocols and Metrics

Results across the field are hard to compare because tasks, metrics and test conditions vary. Table 7 lists the metrics in common use.

Recommendations. Six practices would make results on autonomous platforms more comparable. *(i) Report SNR at the microphones*, measured or controlled, rather than only the source level; performance on drones depends far more on SNR than on method details (Sec. 13.1). *(ii) Separate noise-model data from test data*: when a method learns an ego-noise model, it should be learned from recordings disjoint from the test signals, as in the held-out protocol of our drone case study (Sec. 13.1). *(iii) Fix frame alignment independently of the methods compared*: array and ground-truth frames should be aligned once, on separate data, with a method not under test. *(iv) Report per-condition results*, such as per flight or per SNR, since averages can hide where a method helps most; in our drone case study the benefit of motor-speed information is concentrated in the quiet-source condition. *(v) Report a baseline without a noise model and the noise conditions*, because a noise model can also cost accuracy when the SNR is high and the noise close to diffuse, as on the robot-head array of Sec. 13.4. *(vi) Account for correlated windows*, since neighboring windows of one flight are not independent. Intervals should then come from a block bootstrap and tests from block resampling, as in Sec. 13.1, where the 391 windows count as only about 58 to 282 independent ones.

13. Case Studies on Public Datasets

To test the argument of Sec. 8 rather than assert it, we run reproducible studies with established methods and one shared code base, on three public datasets (Tab. 8). Section 13.1 measures how much an explicit noise model and motor-speed information help in-flight localization on one drone. Section 13.2 repeats this test with recorded speech on a second drone, which was held in place. Section 13.3 asks whether the properties of ego-noise that make such models work hold across five different drones. Section 13.4 applies the same localizers to a robot-head array and a spherical array in a reverberant room at high SNR, to see where a noise model stops helping. The code reproduces every number and figure in this section, and it is available at `https://github.com/Heena-S-Patel/Listening-Through-Ego-Noise-A-Review-of-Acoustic-Perception-with-Reproducible-Case-Studies`. We used Python 3.12, NumPy 2.5, SciPy 1.18 and random seed 0.

13.1. In-flight localization on a drone (DREGON)

DREGON [84] records an 8-microphone array on a quadrotor, together with each rotor's measured speed (logged at

Resource	Platform	Real ego-noise	Ground truth	What it tests
DREGON [84]	drone, 8 mics	yes	motion capture	in-flight localization; per-rotor speed and IMU logged
SP Cup 2019 [20]	drone	yes	yes	localization for search and rescue
SP Cup ego-noise [20]	10 drones, 1–16 mics	yes	–	ego-noise of different drones; used in Sec. 13
AIRA-UAS [74]	drone	yes	–	ego-noise during maneuvers and from nearby drones
AVQ [94]	drone, 8 mics + camera	yes	from video	audio-visual localization, tracking and enhancement outdoors
LOCATA [28, 53]	robot head (12 mics), spherical, planar, hearing aids; static and moving	no	motion capture	localization and tracking of moving talkers in a reverberant room; used in Sec. 13
OVAD [79]	vehicle, 56 mics	partly	yes	detection of occluded vehicles
STARSS23 [82]	static array + 360° video	no	motion capture	audio-visual SELD in real rooms
AudioSet [37]	web video	no	labels	event classification and pretraining
ESC-50 [68]	curated clips	no	labels	environmental sound classification
FSD50K [29]	curated clips	no	labels	open event classification
VGGSound [15]	web video	no	labels	audio-visual recognition
SoundSpaces [9, 13]	simulator	no	exact	audio-visual navigation and rendering
Sonicverse [36]	simulator	no	exact	multisensory household agents
ObjectFolder [35]	object models	no	exact	visual, auditory and tactile object properties
HARK [63]	toolkit	–	–	robot localization, separation, recognition
ODAS [41]	toolkit	–	–	embedded localization, tracking, separation

Table 6. Public datasets, simulators and toolkits. Only DREGON combines real ego-noise with the platform's own motor state and ground-truth directions; none yet combines a moving platform, real ego-noise, platform state, high-rate vibration sensing, multiple sound classes and video.

Task	Metrics	Refs
Localization	angular error (mean, median); accuracy within a threshold, *e.g.* 10°	[28, 42]
Tracking	localization metrics over time; track continuity and swaps	[28]
Event detection	segment- or event-based F-score and error rate	[61]
SELD	location-aware detection plus class-aware localization error	[69]
Separation	SI-SDR	[73]
Speech	word error rate	[71]
Navigation	success rate; success weighted by path length	[9]

Table 7. Common evaluation metrics by task.

about 1 kHz), the flight controller's IMU and the true source direction from motion capture. We use the noise-only flight to learn noise models, two free flights with a white-noise loudspeaker (391 windows of 0.5 s) for evaluation, and the clean tripod recordings mixed with held-out flight noise for a controlled SNR sweep. The noise-only flight lasts 71 s, of which about 60 s have the motors running.

Setup. We compare six localizers on a 3° grid of 7320 directions (120 azimuths and 61 elevations) and a 300–4000 Hz band. (A) SRP-PHAT [23]. (B) SRP-PHAT with tone masking, a simple baseline of our own. It removes the bins at the blade passing harmonics $2kf_0$ of each rotor, where f_0 is the shaft rate from the motor log, since rotor noise contains tones at these frequencies [19]. A bin is removed when it rises at least 3 dB above the local spectral floor, and at most 40% of the bins in the band are removed. (C) MUSIC [78], with each narrowband pseudo-spectrum normalized by its maximum over directions so that no single bin dominates. (D) GEVD-MUSIC (Eq. (3)) with a single noise covariance learned from the noise-only flight [30, 64]. Its diagonal loading is 10^{-3} times the mean noise power, and each generalized eigenvector is scaled to unit norm, since these vectors have no fixed scale. (E) GEVD-MUSIC with six covariances, one per quantile bin of mean motor speed, which is a simple form of state-informed modeling [30, 95]. Each window uses the bin closest to the mean rotor speed logged at its center. (F) SRP-PHAT after a multichannel Wiener filter, which follows the best method of the DREGON paper [84]. Each frame becomes $\hat{\mathbf{x}} = (\mathbf{R}_x - \mathbf{R}_n)\mathbf{R}_x^{-1}\mathbf{x}$, and the generalized eigenvalues of $(\mathbf{R}_x, \mathbf{R}_n)$ are first raised to at least one, which keeps the filter positive semidefinite. F uses the covariance of D, and F with speed bins uses those of E. All methods use a 1024-point STFT with 50% overlap, except B, which uses 4096 points with 75% overlap so that individual

Set	Platform	Mics (aperture)	Motor state	Ground truth
DREGON	MK-Quadro, flying	8 (17 cm)	Rotor speeds	Motion capture
SP Cup	DJI Phantom 4 Pro, flying	8 (6 cm)	None	Noise only
SP Cup	MK Quadro V3, held	8 (15 cm)	None	Noise only
SP Cup	Intel Aero, flying	8 (unpublished)	None	Noise only
SP Cup	Quadrotor (AGH), held	8 (25 cm, measured)	Throttle	Labeled directions
LOCATA	Robot head	12 (12 cm)	None	Optical tracking
LOCATA	Eigenmike	32 (8.4 cm)	None	Optical tracking

Table 8. Datasets used in the case studies. DREGON and SP Cup recordings are sampled at 44.1 kHz and LOCATA at 48 kHz.

rotor tones are resolved. The subspace methods use every second frequency bin. The array and motion-capture frames are aligned once, with method A only, on a separate flight with motors off (median error after alignment 2.5°). This alignment therefore cannot favor any of the compared methods. The motor-speed unit is not documented by the dataset; aligning predicted rotor harmonics with the noise spectrum shows that the logged values are revolutions per second (best-fitting scale 0.997). Localization is scored by the angular error between the estimated and true directions, as the median and as the share of windows within 10°. Neighboring windows are correlated, so we report 95% intervals from a circular block bootstrap with 5 s blocks drawn within each flight, and test paired differences with a block sign-flip test.

These settings are not critical. With diagonal loading between 10^{-5} and 10^{-2}, D stays at 59.8 to 60.4% within 10° and E at 68.0 to 70.1%, whereas 10^{-1} lowers them to 53.5 and 55.5%. Without the unit-norm scaling, D and E reach 54.7 and 59.3%. For F, the matrix order printed in Eq. 2 of the DREGON paper gives 0.3% on our flights, and our order without the eigenvalue limit gives 3.3%. We chose the final form of F after seeing these results, so F may be slightly optimistic.

Localization in flight. Table 9 gives the results. Without a noise model (A, C), most windows are wrong by tens of degrees, since the response peaks on rotor noise arriving from about 45 to 50° above the array. Figure 3 shows this over whole flights. Learning the noise covariance (D) reduces the median error from 76.7° to 6.1°, a gain of 41.9 points [28.6, 54.7] within 10°. Selecting the covariance by motor speed (E) adds 9.5 points [4.9, 13.8] ($p = 0.008$) and reduces the mean error from 31.4° to 22.7°. This gain is concentrated on the quiet-source flight, where E reaches 64.0% against 46.5% for D, a gain of 17.5 points [9.5, 25.5]. On the loud flight it is only 1.0 point [−2.1, 4.7].

How firm the gains are. From the autocorrelation of the paired differences, the 391 windows count as about 282 independent windows for E against D and about 58 for D against A. Blocks of 2.5 and 10 s give intervals of [5.4, 13.8] and [4.1, 14.6] for E over D. An exact McNemar test also

	Error		Within 10°			
Method	Med.	Mean	All	95% CI	Loud	Quiet
A SRP-PHAT	76.7°	72.7°	18.4%	[10.5, 26.9]	36.1%	1.5%
B + tone mask	67.3°	59.5°	33.0%	[23.8, 42.5]	58.6%	8.5%
C MUSIC	72.4°	67.4°	23.3%	[13.0, 34.3]	44.5%	3.0%
D GEVD, one model	6.1°	31.4°	60.4%	[46.0, 73.7]	74.9%	46.5%
E GEVD, speed bins	5.0°	22.7°	69.8%	[55.5, 82.6]	75.9%	64.0%
F Wiener, one model	5.0°	24.0°	71.4%	[59.6, 82.1]	**79.1%**	64.0%
F Wiener, speed bins	**4.9°**	**21.5°**	**72.4%**	[60.1, 83.1]	78.5%	**66.5%**

Table 9. In-flight localization on DREGON, 391 windows of 0.5 s scored against motion capture. Median and mean angular error, and share of windows within 10° over both flights with its 95% block-bootstrap interval, and for the loud-source (191 windows) and quiet-source (200 windows) flights.

favors E, which is correct in 40 windows where D fails and fails in 3 where D is correct ($p < 10^{-8}$), but it treats the windows as independent, so we rely on the block intervals. Two flights are too few to use flights as the unit of a test, so each flight is reported separately, and the gain cannot be generalized to other flights. It also depends on the noise recording. When both models are learned on 35 s spans at five places in the noise-only flight, D reaches 69.1% on average (58.3 to 74.2%) and E 72.0% (67.5 to 73.1%), so most of the gain disappears. Averaging one covariance over the whole flight therefore appears to penalize D. E reaches 64.2 to 69.8% with 2 to 24 speed bins, and six bins give the highest value, so the 69.8% may be slightly optimistic. Spans of 5, 10 and 20 s give E 65.8% (47.8 to 71.6%), 68.5% (58.1 to 73.9%) and 70.7% (62.9 to 73.7%) at the same five places. Short spans are therefore usable on average, but a 5 s span can fail badly.

Motor state or a better filter? The Wiener filter (F) changes the reading of the motor state. With one covariance and no motor data, F reaches 71.4% [59.6, 82.1], which is 52.9 points [42.2, 63.7] above A and 11.0 points [4.9, 17.6] above D ($p = 0.009$). F does not differ measurably from E (1.5 points [−4.6, 8.2], $p = 0.73$). Adding speed bins to F gives 72.4%, a gain of only 1.0 point [−1.3, 3.3] ($p = 0.58$), and 2.5 points [−0.5, 6.0] on the quiet flight. Much of the gain of E over D therefore reflects a weakness of one covariance in GEVD-MUSIC, and not information

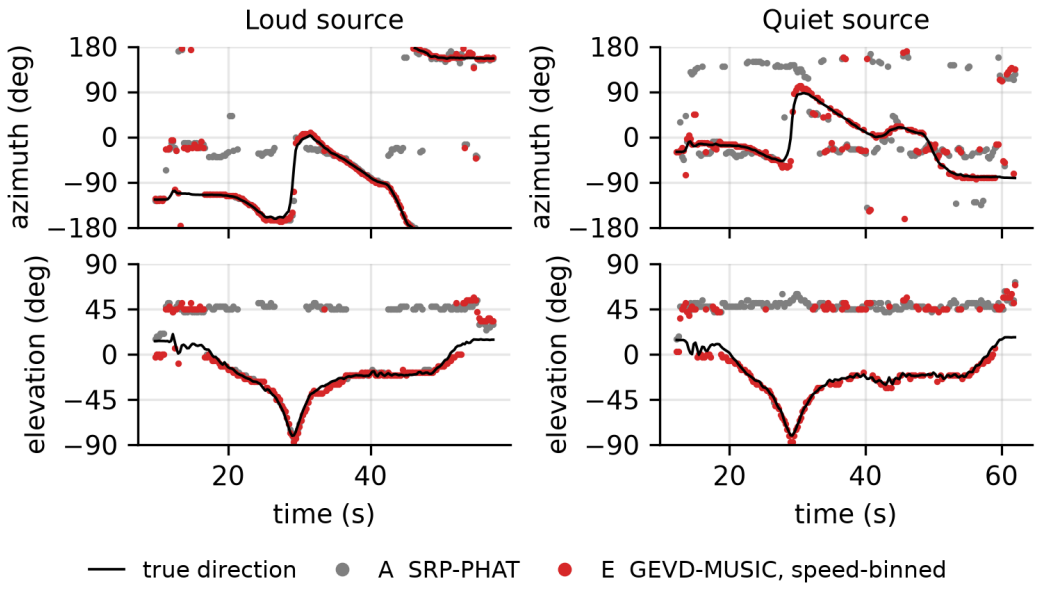


Figure 3. True source direction (black) against the SRP-PHAT baseline (gray) and the speed-binned noise model (red) over two DREGON flights. The baseline locks onto the rotors at about 45–50° elevation; the noise-model method follows the source through the maneuver.

that only the motor state can provide.

Controls for flight. Two controls separate the rotor noise from other effects of flight. On a motors-off recording of the moving array (252 windows, SNR about 32 dB), A and F reach 96.4%, whereas C and D reach only 62 to 64%. C has a fixed elevation bias of −5.1° on this recording, and correcting it raises C to 97.6%. Within 20°, all these methods reach 98 to 100%. Refitting the frames with this bias changes C, D and E on the flights by at most 1.3 points (23.0, 59.1 and 68.8%), so Tab. 9 does not depend on it. Mixing this recording with held-out flight noise tests motion at a controlled SNR. At −10 dB, A, D, E and F reach 67.1, 73.8, 72.6 and 92.1%, and at −15 dB they reach 6.7, 37.3, 46.0 and 62.3%. At −15 dB, E leads D by 8.7 points [4.4, 13.1] ($p = 0.003$) and F leads D by 25.0 points [17.5, 32.5], whereas speed bins lower F by 2.4 points [−4.8, −0.4].

Tone masking. Masking rotor tones (B) raises the share of SRP-PHAT within 10° from 18.4% to 33.0%, but two thirds of the windows still exceed 10° of error. The reason is that much of the noise lies between the tones. In the 1 to 2 kHz band, the predicted blade passing tones cover 24% of the frequency bins but hold only 30% of the noise energy. B uses a 4096-point STFT while A uses 1024 points. With a shared 1024-point STFT, B reaches only 21.2%, so most of its gain over A comes from the finer STFT rather than from the mask. At 4096 points with 75% overlap, as for B, A reaches 28.9%, against 33.0% for B (Tab. 10). The DREGON paper does not state the blade count, so two blades remain an assumption. Its Fig. 3b shows one motor at 80 turns/s with noise peaks near 80, 160, 238 and 316 Hz, with the strongest at 160 Hz, which is consistent with two blades [84]. Combs at one, two and three times the shaft rate give 35.3, 33.0 and 31.5%, so B does not depend strongly on it. The 40% cap on removed bins is never reached, since the mask removes 6.6% of the bins on average and at most 17.7% in any window. Tone masking and noise modeling can also be combined. Removing the masked bins before D and E gives 64.5 and 72.6%, a gain of 4.1 [2.0, 6.4] and 2.8 [0.8, 5.1] points.

Analysis settings. The original papers report each method with different analysis settings. Table 10 therefore repeats the flight experiment with three alternatives, namely a 4096-point STFT for all methods, the frame length, hop, band and frequency weighting reported in the original papers [23, 30, 64, 65], and our STFT with a 300 to 8000 Hz band. MUSIC takes the settings of GEVD-MUSIC [64]. Method E takes those of the state-informed model that inspired it [30], which reports no band, so E keeps ours. SRP-PHAT computes one response per block as in the original [23], and these responses are summed over each window, so that all methods are scored per window. In this column, the generalized eigenvectors are not rescaled, and D reaches 71.1% when the whitened eigenvectors are used instead, as written in [65]. All columns use free-field steering vectors, whereas [30, 64] measured theirs. With the 4096-point STFT, the noise models lose most of their advantage, and raising the overlap

Method	1024 pt 0.3–4 kHz	4096 pt 0.3–4 kHz	Original papers	1024 pt 0.3–8 kHz
A SRP-PHAT	18.4%	28.9%	39.4%	49.1%
B + tone mask	21.2%	33.0%	21.2%	51.2%
C MUSIC	23.3%	33.0%	28.9%	47.6%
D GEVD, one model	60.4%	34.0%	53.2%	78.0%
E GEVD, speed bins	**69.8%**	**38.9%**	**71.9%**	**83.1%**

Table 10. Ablation of the analysis settings on DREGON. Share of windows within 10°. B is our own baseline with no original paper, so that column repeats its 1024-point result.

Method	0 dB	−5 dB	−10 dB	−15 dB	−20 dB	−25 dB
A SRP-PHAT	100%	98%	40%	2%	0%	0%
B + tone mask	100%	100%	46%	2%	0%	0%
C MUSIC	100%	98%	46%	0%	0%	0%
D GEVD, one model	100%	100%	90%	29%	0%	0%
E GEVD, speed bins	100%	100%	90%	35%	0%	0%
F Wiener, one model	100%	100%	**92%**	**46%**	0%	0%
F Wiener, speed bins	100%	100%	90%	**46%**	0%	0%

Table 11. Share of trials within 10° for motors-off DREGON recordings mixed with held-out flight noise (48 mixtures per level).

to 94% does not restore it (D 34.0%, E 38.1%). With its original settings, SRP-PHAT reaches 39.4%, and this gain comes from its band extending to 8 kHz. Restricted to 300 to 4000 Hz, it falls back to 17.6%. When all methods use the wider band, every method improves, and D and E stay 29 and 34 points above A. In every setting, a method with a noise model is the most accurate. Since the source here is white noise, the gain above 4 kHz need not carry over to speech, which carries most of its energy below 4 kHz. F also gains from the wider band, reaching 82.4% [74.4, 89.5], and speed bins bring it to 83.4% [75.4, 90.5], a gain of 1.0 point [0, 2.3] ($p = 0.25$). We therefore keep the 300 to 4000 Hz band, which is similar to those of earlier robot and drone studies [64, 65].

Noise level. For this sweep, 24 motors-off recordings (azimuth 45 to 90°, elevation 0 to -30°) are each mixed with two different flight-noise segments at broadband SNRs from 0 to −25 dB, giving 48 mixtures per level. The noise model is learned on the first half of the noise-only flight (about 26 s with the motors running), and the mixing noise is taken from the second half. The broadband SNR is the ratio of the sample powers of the clean recording and the noise, averaged over all microphones and without band limiting. The frames are aligned in the same way on the motors-off recordings themselves, before any noise is added (median residual 3.0°). Table 11 shows that all methods are accurate at 0 and −5 dB. At −10 dB, D and E still localize 90% of trials within 10°, while the others fall to 40 to 46%. At −15 dB, they keep 29 to 35% against at most 2% for the others, and from −20 dB every method fails. The noise model therefore delays the collapse of accuracy by about 5 dB, but no method reaches the SNR at which distant calls for help would be heard. Resampling the 24 directions, the gain of D over A is 50.0 points [35.4, 64.6] at −10 dB and 27.1 points [8.3, 47.9] at −15 dB. The difference between D and E is 0 at −10 dB and 6.2 points [0, 16.7] at −15 dB (exact McNemar test, $p = 0.25$), so the sweep supports the noise model but not an added benefit of the speed bins. F keeps 92 and 46%, which is 52.1 points [37.5, 66.7] and 43.8 points [29.2, 58.3] above A. F leads D by 2.1 points [−10.4, 16.7] at −10 dB and by 16.7 points [8.3, 27.1] at −15 dB, and speed bins add nothing to F.

Signal quality. Because a beamformer is linear, the target and the noise can be passed through it separately and the output SNR measured exactly. We steer an MVDR beamformer (Eq. (4)) with the learned covariance towards E's *estimated* direction rather than the true one, which reflects realistic operation, and score the SNR gain over a single microphone between 200 and 8000 Hz. Over 48 mixtures at −10 dB input SNR, it raises the SNR by a median of 9.7 dB with the speed-binned covariance and 9.1 dB with a single covariance, against 4.0 dB for delay-and-sum. The median paired gain of speed-binned MVDR is 6.0 dB [5.5, 6.3] over delay-and-sum and 0.6 dB [0.5, 0.7] over one covariance. At −15 dB input, the MVDR gain falls to 4.4 dB, since the estimated direction is often wrong. The diagonal loading is 10^{-2} times the mean noise power, and between 10^{-4} and 10^{-2} the median gain stays at 9.7 to 9.8 dB, while 10^{-1} gives 8.9 dB. This measure suits a white-noise source, whereas speech would also need perceptual metrics. Figure 4 shows that the gain is largest between 1 and 4 kHz.

Rotor tones and the IMU. Rotor tones are predictable from the motor log, but most of the noise energy on this drone is not on those tones. The flight controller's IMU, sampled at about 1 kHz, cannot observe vibration above 500 Hz, which covers the lowest rotor harmonics (from about 140 Hz) but not the band used for localization.

13.2. Recorded speech on a second drone (AGH)

DREGON gives only white noise in flight, so we test speech on the anechoic quadrotor of the SP Cup team from AGH University, Kraków [20]. This set holds white noise from 8 directions and recorded speech from 2 directions with the motors off, and ego-noise at five throttle settings. Its array geometry is not published, so we measured the microphone positions from GCC-PHAT time differences on the white-noise recordings. Each direction is scored with a geometry fitted only to other directions at least 15° away (aperture 25 cm, fit residual 12.5 μs, held-out errors 6.9 to 65 μs). Clean recordings are mixed with noise at throttles of 100, 80 and 60%, at equal SNR between 300 and 4000 Hz,

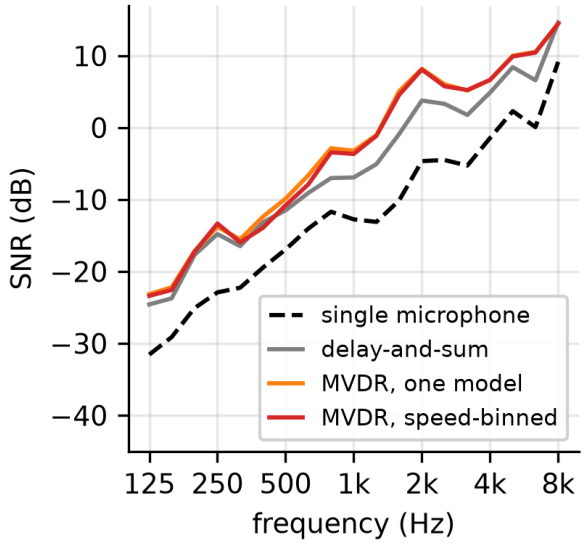


Figure 4. SNR by third-octave band before (single microphone, dashed) and after beamforming towards the estimated direction, averaged over 48 mixtures at −10 dB input SNR. A learned noise covariance (MVDR) roughly doubles the gain of plain delay-and-sum.

with 16 trials per file and throttle. This gives 96 speech and 384 white-noise trials per level. Noise models are learned on the first half of each throttle file, and the mixing noise comes from the second half. D and F pool the throttles, whereas E and F with state use the throttle of the mixture, which is known exactly here.

Table 12 gives the results. At 0 and −5 dB, all methods score 85.4% on speech and, down to −20 dB, 87.5% on white noise. These ceilings come from single directions that stay wrong at every SNR, such as one white-noise direction with an error of 11.9°, which we attribute to the measured geometry. The collapse on speech comes about 10 dB earlier than on white noise. The noise model again extends the working range, with D better than A in 29 of 29 differing speech trials at −15 dB. On this drone the known throttle helps more than on DREGON. E leads D in 18 of 18 differing trials on speech at −20 dB ($p < 0.001$), in 16 of 17 at −25 dB, and in 111 of 117 on white noise at −30 dB. F is not always the best method here, since D beats F in 21 of 21 differing speech trials at −15 dB. These tests treat trials as independent, there are only two speech directions, and the drone is held, so this is supporting evidence and not a flight result. On the two recordings with white noise and running motors together, every method reaches 100% at very high speed (44 windows). At medium speed (50 windows), C reaches 64% and D 98%, whereas A, E and F reach 100%, using the closest throttle setting because these recordings

	Speech				White noise		
Method	−10	−15	−20	−25	−25	−30	−35 dB
A SRP-PHAT	72.9	51.0	9.4	0.0	58.1	0.8	0.0
D GEVD, pooled	**85.4**	81.2	31.2	1.0	**87.5**	52.3	0.5
E GEVD, throttle	**85.4**	**83.3**	50.0	**16.7**	**87.5**	79.7	5.5
F Wiener, pooled	83.3	59.4	25.0	0.0	87.0	37.2	1.6
F Wiener, throttle	**85.4**	79.2	**53.1**	15.6	**87.5**	**80.7**	**8.6**

Table 12. Second drone (AGH, held, anechoic). Share of trials within 10° (%) at equal SNR between 300 and 4000 Hz, with real speech (2 directions, 96 trials per level) or white noise (8 directions, 384 trials per level) mixed with held-out noise. E and F with state use the known throttle.

describe the throttle only in words.

An outdoor flight we could not score. We also tried the outdoor flights of the KumamoTech team in the same collection [20], with 16 microphones and logged position and attitude. The voice in these flights has no activity labels, so we first checked the ground truth on a whistle that is clearly audible. The best method and source reach only 32.0 and 26.4% in azimuth within 20° on the two flights, against 11% by chance, and the alignment differs between flights. The source height is not logged and the position comes from GPS. We therefore do not score these flights.

13.3. Ego-noise across five drones

A single drone cannot show whether these findings are general. We therefore analyze noise-only recordings of four more quadrotors with 8-microphone arrays, released with the IEEE Signal Processing Cup 2019 [20]. These are a DJI Phantom 4 Pro hovering outdoors at 2 and 10 m, an MK Quadro V3 held about 0.5 m above the floor, an Intel Aero in manually controlled hover, and the AGH quadrotor of Sec. 13.2, recorded at varying rotor speeds in an anechoic chamber. Together with 50 s of steady flight from the DREGON noise-only recording, this gives five drones (Tab. 8). The KumamoTech flights could not be validated (Sec. 13.2), and the remaining sets have fewer microphones or could not be downloaded. Because the arrays differ, we use two measures that need no array geometry (Fig. 5). The *tonal share* is the share of 100–4000 Hz noise energy in narrow spectral peaks, at least 6 dB above a 300 Hz running-median floor, computed on 1 s chunks and averaged. It uses 8192-point Welch spectra. As an established check, we also report the harmonic-to-noise ratio (HNR) [5], computed with the autocorrelation method of Praat in the same band and averaged over the microphones. The HNR assumes a single periodic source, whereas a quadrotor has four rotors at slightly different speeds, so we use it only as a second measure of tonality. We define the *learned rejection* to measure spatial structure. It builds on the model of Wang and Cavallaro [92], in which each rotor acts as one directional noise

source and a diffuse part is added. Because the microphones and motors are fixed to the frame, the acoustic mixing is stationary. We learn the four strongest noise directions on the first half of a recording, one per rotor, and remove them from the second half with an orthogonal projection, as in the eigencanceler [45]. The learned rejection is the resulting drop in noise power, averaged over 300–4000 Hz bins. For spatially white noise on eight microphones it is 3 dB, since four of the eight dimensions hold half of the power. For the three arrays whose geometry was published, we compare it with a spherically diffuse noise field on the same array. This reference uses the theoretical coherence $\sin(2\pi fd/c)/(2\pi fd/c)$ between microphones at distance d, not a simulation. Since the rejection depends on the aperture and microphone spacing, values are compared with this reference and not directly between arrays.

Results. Tonality differs by a factor of up to 60 between drones, from 1.2% of the noise energy on the Intel Aero, through 15.7% on DREGON, to 69.7% on the MK Quadro V3. However, the two highest values come from drones that were not in self-powered flight. They are marked with an asterisk (*) in Fig. 5. Among the three flying drones, the tonal share differs by a factor of about 13. Only three flying drones remain, so this is weak evidence about the potential of tone-based methods. The two held drones rank highest in all 11 estimator settings we tried, with peak thresholds of 3 to 10 dB, median widths of 150 to 600 Hz and 4096 to 16384 points. The absolute values depend strongly on these settings, with DREGON ranging from 0.6 to 64.6%, so only this ordering should be relied on. The HNR ranks the five drones similarly (Spearman $\rho = 0.8$). It is also highest on the two drones that were not flying, the anechoic quadrotor (4.8 dB) and the MK Quadro V3 (2.6 dB), and lowest on the DJI Phantom 4 Pro (0.0 dB). In contrast, a learned spatial subspace removes 6.0 to 12.9 dB on every drone, well above the 3 dB expected for white noise, and the model stays valid from one half of a recording to the other. The rejection grows by about 1 to 4 dB per added eigenvector up to six, with no knee at four, so four follows from the noise model and is not selected by the data. With 1 to 6 eigenvectors, it stays above the diffuse reference on the DREGON and MK Quadro arrays and below it on the DJI Phantom array. On the DREGON and MK Quadro V3 arrays, with apertures of 17 and 15 cm, it removes 4.5 and 2.0 dB more than it would from diffuse noise, so the rotor noise is spatially more structured than background noise. The MK Quadro recordings have one silent channel, so its diffuse reference uses the other seven microphones, for which white noise would give 3.7 dB. On the 6 cm DJI Phantom array, however, a spatial model removes *less* ego-noise than it would remove from diffuse noise (8.2 against 19.0 dB). We checked this directly with the magnitude-squared coherence between microphones, which is 0.25 on average (0.14 below 1 kHz), against 0.51 (0.92) for diffuse noise. The noise is therefore much less correlated than a diffuse field, although not uncorrelated, since the rejection still exceeds 3 dB. Airflow over each microphone capsule may cause this, but our data cannot confirm it. Spatial models cannot remove such noise; larger apertures, flow shielding or single-channel suppression are needed instead. The measured AGH geometry adds a third large array. With an aperture of 25 cm, its diffuse reference is 4.9 dB, so its rejection of 12.9 dB exceeds that of diffuse noise by 8.1 dB, and it stays above the reference for 1 to 6 eigenvectors (5.2 to 19.5 dB against 1.2 to 8.6 dB). Per throttle setting, it ranges from 10.0 dB at full throttle to 15.1 dB at 40%, so the spatial structure is present at every speed we could test. This geometry is measured rather than published, so it is weaker evidence than the other two arrays.

Flight condition and tonality. In flight, the controller continually adjusts the rotor speeds. This smears the rotor harmonics over each analysis window. We measured the rotor-speed variation in each 1 s window of the DREGON noise-only flight as the standard deviation of the logged speeds. Windows with more variation have a lower tonal share (Spearman $\rho = -0.34$, $p = 0.016$, 50 windows). At the same mean rotor speed, their harmonic peaks are also lower (Fig. 6). Tonality measured on a drone that is not flying can therefore overstate the tonality in flight.

13.4. A robot-head and a spherical array in a reverberant room (LOCATA)

To test the localizers on a different platform and array, we use the robot-head array of the LOCATA development set [28, 53]: a prototype NAO head with 12 microphones in a pseudo-spherical arrangement, recorded in a room with a reverberation time of about 0.5 s, with optical tracking of the array and the sources and voice activity labels for every sample. We use the three single-source tasks: a static loudspeaker with a static array (task 1), a moving talker with a static array (task 3), and a moving talker with a moving array (task 5). The ground-truth direction is computed in the array's own frame from the tracked pose, so no alignment is fitted, and windows are scored when the source is active in at least 80% of their samples (396 windows in total). Methods B and E need motor speeds, which this platform lacks; method D learns its noise covariance from frames of the same recording in which the source is silent, at least 0.5 s from any speech so that the reverberant tail does not leak in (or from all silent frames when fewer than 20 such frames exist). This happens for two recordings of task 1 (15 and 8 frames), and the other recordings give 29 to 1396 noise frames. Method F uses the same covariance as D. To test the result further, we add the 32-microphone spherical Eigenmike (aperture 8.4 cm) in the same room, and the LOCATA

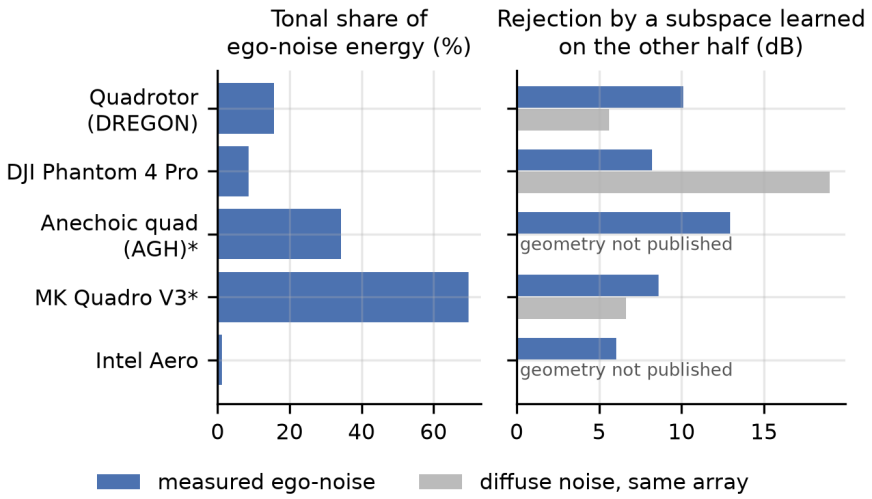


Figure 5. Ego-noise of five quadrotors. Left: share of noise energy in rotor tones. Right: noise power removed from one half of a recording by a four-dimensional spatial subspace learned on the other half, against a diffuse noise field on the same array where the geometry is published. * represents a drone that was not flying during the recording. The MK Quadro V3 was held about 0.5 m above the floor. The AGH quadrotor stood still in an anechoic chamber. Its geometry was measured by us (Sec. 13.2), so the figure shows no diffuse reference for it.

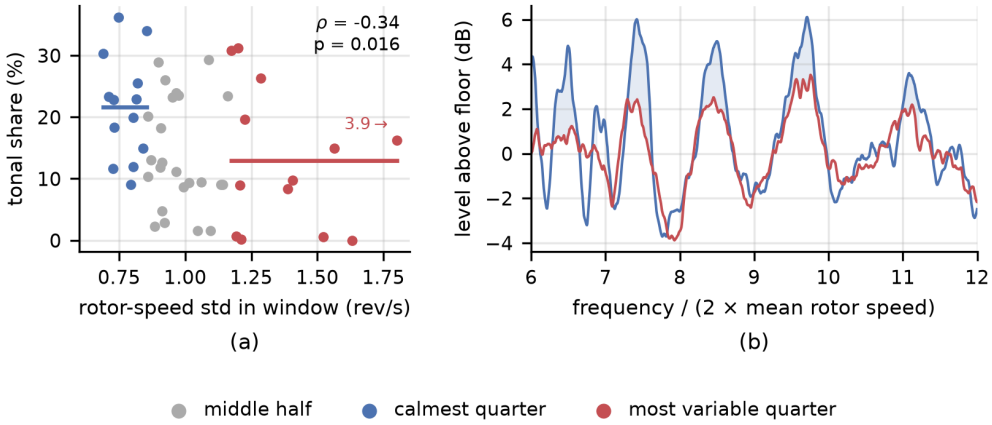


Figure 6. Rotor-speed variation and tonal share in the DREGON noise-only flight (50 windows of 1 s). (a) Tonal share against the standard deviation of the rotor speeds within each window. The standard deviation is averaged over the four rotors. Horizontal lines mark the mean of the calmest quarter (21.6%) and of the most variable quarter (12.9%). One window with 3.9 rev/s is drawn at the edge. (b) Mean spectrum of these two quarters. Each window is aligned to its own mean rotor speed. Both quarters have the same mean speed (80.7 and 80.8 rev/s). The harmonic peaks are lower in the variable windows (shaded).

evaluation set (884 windows from 13, 5 and 5 recordings of tasks 1, 3 and 5), on which nothing was tuned. The protocol is the same.

Results. Table 13 shows that the classical localizers are accurate on this array: SRP-PHAT places 91 to 100% of windows within 10°, with median errors of 2.8 to 4.8°, also when both the talker and the array move. The noise model does not help here; with moving talkers it lowers the share within 10° by 18 to 33 points relative to SRP-PHAT, whether or not the reverberant tail is excluded. The diagnostics explain why. The SNR at the array is high, 16 to 25 dB across recordings, and the background noise is close to diffuse: 75 to 89% of its energy lies in the four strongest spatial components, against 79% for a diffuse field on the same array. There is little structured noise to remove, so whitening with a covariance estimated from a few hundred frames of changing background noise mainly distorts the spatial spectrum. Four controls support this. First, D falls below MUSIC,

Method	Task 1 (34)	Task 3 (151)	Task 5 (211)
A SRP-PHAT	100% / 4.8°	91.4% / **2.8°**	**99.1%** / 2.8°
C MUSIC	100% / 3.1°	86.1% / 3.2°	95.7% / 2.8°
D GEVD, 0.5 s guard	100% / 3.2°	69.5% / 5.4°	68.2% / 4.0°
D GEVD, no guard	100% / **2.8°**	58.3% / 8.0°	81.0% / 4.6°
F Wiener, 0.5 s guard	100% / 3.8°	**92.7%** / **2.8°**	**99.1%** / **2.6°**

Table 13. Localization with the LOCATA robot-head array: share of windows within 10° / median angular error. Window counts in parentheses. Task 1: static loudspeaker and array; task 3: moving talker; task 5: moving talker and moving array.

which isolates the whitening, by 16.6 points [4.3, 30.2] on task 3 and 27.5 points [3.1, 38.8] on task 5, resampling the recordings. Second, raising the loading from 10^{-3} to 10 brings D back to 85.4 and 96.7%, close to MUSIC at 86.1 and 95.7%, whereas loading up to 10^{-2} changes little. Third, reverberant tails do leak into unguarded frames, which are up to 8 dB louder, but they are not the main cause. With guards of 0.25, 0.5 and 1 s, D reaches 70.2, 69.5 and 67.5% on task 3 and 84.8, 68.2 and 66.8% on task 5. Fourth, a covariance from the 100 nearest silent frames (about 1 s) raises task 5 only to 74.9%, so array motion explains part of the loss at most. The LOCATA papers do not state whether the head was powered [28, 53]. On task 5, silent frames are about 9 dB louder and more tonal than on task 3 (tonal share 42 to 64% against 1 to 8%), so the moving setup adds noise of its own and this array is not free of platform noise.

Held-out recordings and a second array. Table 14 tests the result on data not used so far, and only partly confirms it. On the held-out recordings of the robot head, D stays below MUSIC by 13.6 points [6.4, 20.8] on task 3, but by only 0.4 points [−4.2, 5.2] on task 5, so the large task 5 loss of the development set does not replicate. On the Eigenmike, D falls below MUSIC by 51.7 points [31.1, 80.0] and 42.1 points [21.2, 57.6] on task 5 of the two sets, and by 10.6 points [6.5, 18.6] and 3.0 points [−4.9, 14.7] on task 3. Six of the eight comparisons therefore show a loss, but its size varies widely between recordings and arrays. On this array, every method has a median error of about 8° on the static task 1, and all reach 100% within 15°. This suggests a fixed offset in the array frame that we did not correct, so the table gives both thresholds, and the loss holds at both. The Wiener filter separates the covariance from the whitening. With the same noise covariance as D, F stays within 2 points of SRP-PHAT in all these comparisons (Tabs. 13 and 14), for example −1.7 points [−3.7, 0] on task 3 of the held-out robot-head recordings and −1.1 points [−2.1, 0.3] on task 5 of the held-out Eigenmike recordings. The loss therefore comes from whitening inside a subspace method, not from using a noise covariance as such.

13.5. What the case studies show

Taken together, the studies support a qualified version of the argument of Sec. 8. *(i)* When ego-noise dominates and is spatially structured, an explicit noise model is the largest single gain. On DREGON it raised in-flight accuracy within 10° from 18% to 60%, and to 70% with speed bins (Tab. 9), it delayed the collapse of accuracy by about 5 dB (Tab. 11), and it doubled the beamforming gain (Fig. 4). Spatial noise models removed 6 to 13 dB of ego-noise on all five drones (Fig. 5). In every analysis setting we tried, a method with a noise model was the most accurate (Tab. 10). *(ii)* The added benefit of the motor state is smaller and less general. With GEVD-MUSIC on DREGON, speed bins gained 17.5 points on the quiet-source flight, but a Wiener filter with one covariance matched this accuracy without any motor data, and the gain shrank when the single covariance was learned on a shorter span. On the held AGH drone with a known throttle, the state added 18.8 points on speech at −20 dB, close to the collapse (Tab. 12). *(iii)* The value of tone-based methods depends on the drone and on whether it is flying. Tonality ranged from 1% to 70% of the ego-noise energy, and rotor-speed variation in flight lowered it(Figs. 5 and 6). *(iv)* A noise model is not free. When the SNR is high and the background close to diffuse, as on the robot head, GEVD-MUSIC lost accuracy against MUSIC in six of eight comparisons on two arrays and two LOCATA sets, by very different amounts (Tabs. 13 and 14). A Wiener filter with the same covariance stayed within 2 points of SRP-PHAT, so the failure belongs to whitening in a subspace method. *(v)* Array aperture and airflow limit what any spatial model can do, as the 6 cm array showed, where the noise was only weakly correlated between microphones. *(vi)* Recorded speech on the AGH drone collapsed about 10 dB earlier than white noise, so white-noise results overstate what a drone can hear. All methods in these studies are established; the contribution is the controlled, reproducible comparison across platforms.

Practical guidelines. (1) Record a short noise-only segment and learn a spatial noise model. In our data, 20 s with the motors running gave about the same accuracy as longer recordings, whereas 5 s spans were unreliable. A Wiener filter was a robust first choice, although GEVD-MUSIC was better on AGH speech at −15 dB. (2) Measure the tonal share of the noise in flight before investing in tone masking. (3) Consider larger arrays and shield the microphones from airflow. In our data, only the three arrays of 15 cm or more, one with a measured geometry, showed ego-noise more structured than diffuse noise, against one smaller array, so this is an observation rather than a general rule. (4) Check the SNR and how diffuse the background is before applying a noise model, and always report a baseline without it. At high SNR, prefer a filter with the noise covariance over whitening in a subspace method.

	Robot head, evaluation set			Eigenmike, development set		Eigenmike, evaluation set	
Method	Task 1 (173)	Task 3 (236)	Task 5 (475)	Task 3 (151)	Task 5 (211)	Task 3 (236)	Task 5 (475)
A SRP-PHAT	100	90.7	98.1	57.0 / 78.1	74.9 / 94.3	60.2 / 78.0	66.7 / 94.5
C MUSIC	100	90.3	94.7	45.7 / 70.9	71.6 / 90.0	47.9 / 73.7	65.1 / 90.9
D GEVD-MUSIC	100	76.7	94.3	35.1 / 56.3	19.9 / 19.9	44.9 / 61.4	22.9 / 30.9
F Wiener	100	89.0	98.1	56.3 / 80.1	74.4 / 93.8	61.0 / 78.8	65.7 / 93.9

Table 14. LOCATA beyond Tab. 13. Share of windows (%) within 10° for the robot head, and within 10° / 15° for the Eigenmike, with window counts in parentheses. The evaluation set is held out, and nothing was tuned on it.

Threats to validity. Our studies have six main limitations. First, the target sources in the DREGON experiments are loudspeakers playing white noise, which is easier to localize than speech or calls, and recorded speech comes only from a held drone with two directions. Second, only DREGON provides rotor speeds, and AGH provides only throttle settings on a held drone, so the benefit of state information in flight is shown on a single drone. Third, the LOCATA results come from two arrays in one room, so they should be confirmed in other settings. Fourth, with 48 trials per SNR level, differences of a few points in the controlled SNR experiment are within noise. Fifth, DREGON provides only two scored flights, so flight-level variation cannot be estimated. Sixth, the form of F was chosen after seeing flight results, the AGH geometry is measured rather than published, and the Eigenmike results include an uncorrected offset.

14. Open Challenges and Research Directions

1. Hearing below −15 dB. On drones, even noise-model methods fail between −15 and −20 dB (Sec. 13.1), yet distant calls for help fall in exactly this range. Recorded speech on a held drone failed about 10 dB earlier than white noise (Sec. 13.2), and drone recordings can reach SNRs as low as −57 dB [44]. Smooth, physically grounded noise models that vary with each rotor's speed [95], combined with learned suppression [43, 93], are the most promising route. Hybrid methods that first remove tones predicted from the motor state and then whiten the remaining spatial noise exploit complementary structure. Longer integration windows help when the target is persistent, but conflict with the motion of the platform; tracking with motion compensation (direction 5) can bridge the two.

2. From masking to cancellation with vibration references. Motor state gives the frequencies of rotor and motor tones but not their phase. A high-rate vibration sensor near each motor, combined with order tracking, could supply a phase-coherent reference and allow tones to be subtracted rather than masked. Standard flight-controller IMUs are sampled too slowly for this (Sec. 13.1), so dedicated sensors are needed; to our knowledge this direction is largely unexplored on autonomous platforms. A first study predicts drone ego-noise from propeller vibration with a recurrent network and removes it by spectral subtraction [83]. With an accelerometer sampled at 2.5 kHz, it predicts only the noise power, not the phase of the tones. Its value depends on the drone: the tonal share of ego-noise ranged from 1% to 70% across the five drones of Sec. 13.3. Rotor-speed variation in flight also smears the tones. Tonality should therefore be measured in flight before tone cancellation is pursued.

3. Transfer across platforms. Ego-noise models are platform-specific, and learned models trained on one drone or robot rarely transfer. Our multi-drone analysis suggests a starting point: on every drone, a spatial noise model learned from the first half of a recording of tens of seconds remained valid on the second half (Sec. 13.3), so short calibration flights may suffice. Self-supervised adaptation from such recordings, and models conditioned on platform geometry and state rather than identity, would reduce the cost of deploying audition on a new platform. Array design matters as much as the algorithm: on a 6 cm array a spatial model removed less ego-noise than it would remove from diffuse noise. Large pretrained audio encoders [16, 38, 50] are a natural starting point if they can be made robust to ego-noise.

4. Simulation with ego-noise. Embodied simulators render room acoustics well [13, 36, 54] but omit the agent's own noise. Adding parametric ego-noise driven by the simulated platform's state, calibrated on real recordings such as DREGON or AIRA-UAS [74, 84], would make sim-to-real transfer of audio-visual policies far more credible.

5. Temporal reasoning, active listening and confidence. A platform should also decide *which* method to trust. In our studies a noise model was essential on a drone, but whitening with it in GEVD-MUSIC cost accuracy on a quiet robot head and a spherical array, whereas a Wiener filter with the same covariance did not (Sec. 13.4). Selecting or weighting methods from the measured SNR and noise structure is therefore a practical first step. Most localizers decide each short window independently. Tracking with the platform's own motion compensated [27, 28, 89] and calibrated confidence from the sharpness of the spatial spectrum would let

an agent know when it is wrong. An agent that knows it is uncertain can also act to listen better, by moving [57, 62] or, on a drone, by briefly changing its rotor speeds.

6. Open-vocabulary hearing. Audio-language models [21, 25, 39, 86] could let a platform report what it hears in words and search for sounds described in language. Their robustness to ego-noise, their latency and their energy cost on embedded hardware have yet to be measured.

7. Benchmarks that match the problem. A shared benchmark with a moving platform, real ego-noise, platform state, high-rate vibration sensing, realistic intermittent sources (calls, whistles, speech) and video (Sec. 11), evaluated with the protocols of Sec. 12 and archived with a persistent identifier, would align robot audition, drone audition and embodied learning around one problem.

8. Efficiency and energy. Few drone localization systems run in real time on board [17]. Method papers should report computational cost on embedded hardware alongside accuracy, as lightweight localization work already does [40].

9. Privacy and responsible use. Always-on microphones on robots, drones and vehicles can record people who have not consented. Processing on board, keeping only the outputs a task needs (events and directions rather than raw audio), and suppressing speech content when it is not needed are practical safeguards that systems and benchmarks should support.

15. Conclusion

Acoustic perception gives autonomous systems a sense that reaches around corners and through occlusion, and the methods to exploit it, from array processing to large pretrained, audio-visual and language-grounded models, are increasingly mature. What holds hearing back on real platforms is the platform itself. This review has argued that ego-noise is the defining problem of the field, and that the platform's own knowledge of its state is the key to solving it. Across robots and drones, methods that model ego-noise explicitly and use motor or joint data outperform those that treat it as generic interference when ego-noise dominates. Our case studies on three public datasets quantify this. A learned noise model raised in-flight accuracy within 10° from 18% to 60%, and to 70% with motor-speed bins, and spatial noise models removed 6 to 13 dB of ego-noise on all five drones studied. The motor state helped most when the target was quiet, but a Wiener filter with one noise model reached 71% without motor data, so its added value is smaller than it first appears. The value of tone-based methods varies widely between drones and flight conditions, recorded speech failed about 10 dB earlier than white noise, and whitening with a noise model costs accuracy where there is little structured noise to remove. Closing the remaining gap will require bringing together three communities that have so far worked largely apart: robot and drone audition, which understands ego-noise; audio machine learning, which provides robust representations; and embodied vision and learning, which provides the tasks, simulators and benchmarks that turn hearing into action.

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